

HRMS – Company Holiday & Leave Calculation

Complete QA Test Case & Regression Test Document

Purpose: Simple, detailed test cases for developers, QA testers, HR users and business analysts.

Implementation status: Coding complete. This document covers functional, negative, edge-case and regression testing.

Base URL: <http://localhost:5168/>

1. Test Environment & Common Data

CompanyId: 023FF1F8-4DD0-4942-1955-08DEFD16782B

FY2027 LeaveYearId: 8BB8836A-3920-4DA2-41CF-08DEFD305CC8

Period: 2027-04-01 to 2028-03-31

EmployeeId: 19ED1DF1-C3CC-4614-85F0-08DEF7B598A7

LeaveTypeId: 9A5FF5C9-49DF-43F4-5107-08DEFD337F69

Full-day DayPart: 30000000-0000-0000-0000-000000000001

Half-day DayPart: 30000000-0000-0000-0000-000000000002

Existing Holiday 1: 58400962-69FE-4253-BBA4-08DEFED654D1 → 2027-04-23

Existing Holiday 2: FFC4F188-EAC3-4B38-0AB3-08DEFF4B3B2E → 2027-04-30

2. How to Read a Test Case

For every test: prepare the data, call the endpoint or perform the HR action, check the API response, and finally verify the database when the test affects leave calculation.

PASS: Actual result matches expected result.

FAIL: Actual result differs from expected result.

BLOCKED: Test cannot be executed because required data or endpoint is unavailable.

3. Company Holiday CRUD Tests

TC-CH-001 — Create valid company holiday — Positive

POST /api/company-holidays

```
{ "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8", "holidayDate": "2027-05-10", "name": "Company Annual Day", "holidayType": "COMPANY", "isOptional": false }
```

Expected: HTTP 200/201, new holiday ID returned, IsActive=true and IsDeleted=false.

TC-CH-002 — Invalid LeaveYear — Negative

```
{ "leaveYearId": "00000000-0000-0000-0000-000000000000", "holidayDate": "2027-05-10", "name": "Test Holiday", "holidayType": "COMPANY", "isOptional": false }
```

Expected: request rejected; no holiday created.

TC-CH-003 — Duplicate holiday date in same LeaveYear — Negative

Use existing 2027-04-23 and attempt to create another holiday for the same LeaveYear and date.

Expected: request rejected. Database unique index LeaveYearId + HolidayDate prevents duplicate dates.

TC-CH-004 — Same date in different LeaveYears — Positive

Create the same calendar date in two different LeaveYears.

Expected: allowed because uniqueness is per LeaveYear, not globally per date.

TC-CH-005 — Empty holiday name — Negative

```
{ "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8", "holidayDate": "2027-05-15", "name": "",  
  "holidayType": "COMPANY", "isOptional": false }
```

Expected: validation error and no record created.

TC-CH-006 — Holiday outside LeaveYear — Negative / Verify Rule

FY2027 ends 2028-03-31. Try a holiday on 2028-04-01.

Expected: reject if LeaveYear date-range validation exists. This must be confirmed against the current validator because the current implementation evidence does not explicitly establish this rule.

TC-CH-007 — Optional holiday — Positive

```
{ "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8", "holidayDate": "2027-05-20", "name": "Optional  
Holiday", "holidayType": "OPTIONAL", "isOptional": true }
```

Expected: holiday created. Current leave calculation does not automatically deduct it.

TC-CH-008 — Inactive holiday — Positive

Set IsActive=false.

Expected: holiday remains stored but does not reduce leave days.

TC-CH-009 — Get all holidays — Positive

GET /api/company-holidays

```
Expected response structure: { "items": [], "totalRecords": 0, "pageNumber": 1, "pageSize": 10,  
  "totalPages": 0 }
```

Expected: actual records and pagination metadata are returned.

TC-CH-010 — Get holiday by ID — Positive

GET /api/company-holidays/{holidayId}

Expected: correct holiday returned.

TC-CH-011 — Get unknown holiday — Negative

GET /api/company-holidays/00000000-0000-0000-0000-000000000000

Expected: standard 404/not-found response.

TC-CH-012 — Update holiday — Positive

PUT /api/company-holidays/{id}; change a valid holiday date/name.

Expected: record updated successfully.

TC-CH-013 — Update holiday to duplicate date — Negative

Change one holiday to a date already used by another holiday in the same LeaveYear.

Expected: request rejected by business/database uniqueness rule.

TC-CH-014 — Delete holiday — Positive

DELETE /api/company-holidays/{id}

Expected: soft-delete behavior: IsDeleted=true, DeletedOn populated, and deleted holiday is not used in calculation.

4. Basic Leave Calculation Tests

TC-LC-001 — Normal leave without holiday — Positive

Leave: 2027-05-05 → 2027-05-07. No holiday.

Expected: 1 + 1 + 1 = 3.00 days.

TC-LC-002 — One holiday in the middle — Positive / PASS already tested

Leave: 2027-04-28 → 2027-05-02. Holiday: 2027-04-30.

28 = 1 29 = 1 30 = 0 (holiday) 01 = 1 02 = 1 Total = 4.00

Expected: 4.00. This scenario has already passed in the current project.

TC-LC-003 — Holiday on first day — Edge Case

Holiday: 2027-05-10. Leave: 2027-05-10 → 2027-05-12.

Expected business result: 2.00 if the first day is a full company holiday. Verify actual implementation result.

TC-LC-004 — Holiday on last day — Edge Case

Holiday: 2027-05-12. Leave: 2027-05-10 → 2027-05-12.

Expected business result: 2.00 if the last day is a full company holiday. Verify actual implementation result.

TC-LC-005 — Single-day leave on holiday — Edge Case

Holiday: 2027-05-10. Leave: 2027-05-10 → 2027-05-10.

Expected: 0.00 for a full-day company holiday. Verify actual implementation result.

TC-LC-006 — Single normal day — Regression

Leave: 2027-05-11 → 2027-05-11. No holiday.

Expected: 1.00.

TC-LC-007 — Multiple holidays — Edge Case

Holidays: May 10, May 12, May 14. Leave: May 10 → May 14.

Expected: 0 + 1 + 0 + 1 + 0 = 2.00.

TC-LC-008 — Optional holiday — Negative/Business Rule

Optional holiday: 2027-05-10. Leave: May 10 → May 12.

Expected under current implementation: 3.00 because IsOptional=true is excluded from automatic deduction.

TC-LC-009 — Inactive holiday — Negative/Business Rule

Holiday: May 10, IsActive=false. Leave: May 10 → May 12.

Expected: 3.00.

TC-LC-010 — Deleted holiday — Negative/Business Rule

Holiday: May 10, IsDeleted=true. Leave: May 10 → May 12.

Expected: 3.00.

TC-LC-011 — Holiday belongs to another LeaveYear — Regression

Create/use a holiday in FY2026 for a date also present in FY2027 leave.

Expected: FY2026 holiday must not reduce FY2027 leave. Expected 3.00 for a three-day request.

5. Create / Submit / Update Leave Tests

TC-LR-001 — Create normal leave — Regression

Create DRAFT leave for three normal days.

Expected: DRAFT and TotalDays=3.00.

TC-LR-002 — Create leave containing company holiday — PASS already tested

Leave 2027-04-21 → 2027-04-25; holiday 2027-04-23.

Expected: DRAFT and TotalDays=4.00. Already passed.

TC-LR-003 — Submit leave containing holiday — PASS already tested

Submit DAAA731C-86A6-4F2F-1EB4-08DEFF48E482.

Expected: PENDING and TotalDays=4.00. Already passed.

TC-LR-004 — Holiday added after leave creation — Important

Create leave for three days with no holiday. Add a company holiday for the middle day. Submit the leave.

Expected: Submit recalculates and reduces 3.00 to 2.00.

TC-LR-005 — Update draft containing holiday — PASS already tested

A6562C90-4F07-49D9-B1AF-08DEFED676EE was updated to 2027-04-28 → 2027-05-02 with holiday 2027-04-30.

Expected/actual: TotalDays=4.00. Already passed.

TC-LR-006 — Update pending leave — Negative / PASS already tested

Attempt to update a PENDING leave.

Expected: { "StatusCode": 409, "Message": "Only draft leave requests can be updated." }

TC-LR-007 — Overlapping leave — Negative / Regression

Create a leave that overlaps an existing PENDING or APPROVED request.

Expected: 409 Conflict with the existing overlap message. This rule must continue working after the calculation change.

6. Day-Part Tests

TC-DP-001 — Full-day leave — Regression

One normal date with full-day start/end parts.

Expected: 1.00.

TC-DP-002 — Half-day leave — Regression

One normal date with half-day start/end parts.

Expected: 0.50.

TC-DP-003 — Half-day on company holiday — Important Edge Case

Request a half-day leave on a full company holiday.

Expected: confirm the HR business decision and then verify implementation. This case must not be assumed without testing.

7. API Validation Negative Tests

TC-API-001 — Invalid GUID. Send leaveYearId='ABC'. Expected 400 validation error.

TC-API-002 — Missing LeaveYearId. Expected 400 validation error.

TC-API-003 — Invalid date format. Expected 400 validation error.

TC-API-004 — Missing holiday name. Expected validation error.

TC-API-005 — Invalid route ID / request ID mismatch. Expected application's route validation error.

8. Pagination Regression

Create more than 10 holidays. Call GET /api/company-holidays?pageNumber=1&pageSize=10.

Expected: correct totalRecords, pageSize=10, and totalPages. Page 2 must contain the remaining records.

9. Full Regression Suite

Area	Regression Test	Expected
Leave creation	Normal full-day leave	PASS
Leave creation	Half-day leave	PASS
Leave validation	Invalid employee	Rejected
Leave validation	Invalid LeaveYear	Rejected
Overlap	Pending overlap	409
Overlap	Approved overlap	409
Submit	Draft → Pending	PASS
Approve	Pending → Approved	PASS
Reject	Pending → Rejected	PASS
Cancel	Existing cancellation workflow	PASS
Update	Draft update	PASS
Update	Pending update	409
Get	Leave request list	PASS
Pagination	Leave request pagination	PASS
Day parts	Full/half day calculation	PASS
Holiday	Middle holiday deduction	4.00 in proven scenario

Area	Regression Test	Expected
Holiday	Optional/inactive/deleted	Not deducted

10. Priority Test Order

Run these first: P01 normal leave; P02 middle holiday; P03 first-day holiday; P04 last-day holiday; P05 single-day holiday; P06 multiple holidays; P07 optional holiday; P08 inactive holiday; P09 deleted holiday; P10 wrong LeaveYear; P11 Create with holiday; P12 Submit with holiday; P13 holiday added after Create then Submit; P14 Update draft with holiday; P15 overlap regression.

11. Current Tests Already Proven

Company Holiday CRUD is implemented and tested. Create Leave with a middle company holiday is proven. Submit Leave with the same holiday is proven. Update Leave with a different company holiday is proven. Existing overlap validation and pending-update protection were also observed working.

Proven calculation 1: 2027-04-21 → 2027-04-25 with 2027-04-23 holiday = 4.00 days.

Proven calculation 2: 2027-04-28 → 2027-05-02 with 2027-04-30 holiday = 4.00 days.

12. Final QA Sign-Off

The feature should be marked PASS/Ready for Commit only after the remaining edge cases and regression tests are executed. A test failure should first be classified as either a real defect, an existing business rule, or an unconfirmed business requirement before changing code.

Recommended final status: Coding complete; QA testing in progress.

End of document.